Hongye Xu

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Expected graduation: 10/2027

Summary

Ph.D. candidate in **Imaging Science (Computer Vision / Machine Learning)** at Rochester Institute of Technology (RIT) working on **vision foundation models, multimodal LLMs/VLMs, and continual learning**—designing, training, and evaluating deep models on large-scale real-world data, and adapting them to new tasks under tight parameter and compute budgets. Currently a **Research Intern on the Perception team at Pony.ai**, distilling vision foundation models into compact onboard backbones for **autonomous-driving perception**. First-author papers at **ECCV**, **ICLR**, and **CVPR (Findings)**, co-first-author work in *Briefings in Bioinformatics*, and further publications at **ICCV** and **GLSVLSI**; **ICML 2026 Gold Reviewer**. Hands-on with large vision-language models (CLIP, BLIP, LLaVA, Qwen), **knowledge distillation**, parameter-efficient fine-tuning (LoRA, MoE-LoRA), **training-free inference-time methods**, and large-scale distributed-training / benchmarking pipelines (∼3M instances; test sets up to 214k samples).

Education

Rochester Institute of Technology — Aug. 2021 – Present
Ph.D. Candidate, Imaging Science (Computer Vision / Machine Learning) — *Rochester, NY, USA*

Rutgers University — Sep. 2018 – Jan. 2021
M.S. in Statistics (Data Science), GPA 3.95/4.00 — *New Brunswick, NJ, USA*

Jilin University — Sep. 2011 – Jul. 2015
B.S. in Mathematics & Applied Mathematics — *Changchun, Jilin, China*

Industry Experience

Pony.ai — Aug. 2026 – Present
Research Intern, Perception — *Fremont, CA, USA*

Vision Foundation Model Distillation for Autonomous Driving

- Distilled a frozen **DINOv3 ViT-7B** teacher into a compact **ViT-S+** backbone for camera-based **BEV perception** using KL-based global-feature and intra-image patch-relation distillation. Trained on mixed driving and web data, with weight interpolation against the original backbone to preserve generality.
- Improved **BEV 3D detection by 2.4 AP** and **front-camera 2D detection by 1.3 mAP** over the original DINOv3 ViT-S+ backbone.

Vision–Language Distillation for Long-Tail Pedestrian Classification

- Designed offline distillation from a patch–text-aligned **TIPSv2** teacher to a lightweight onboard student, combining embedding alignment, open-set prompt-score distillation, per-prompt calibration, and gradient normalization. Defined teacher-first, in-domain validation gates for model adoption.
- Reduced **false positives on rare pedestrian attributes by 58%** at matched recall on a held-out long-tailed pedestrian test set, with no regression on existing closed-set classes.

Research Experience (Selected)

Training-Free Routing Repair for Multimodal LLM Continual Learning — Jun. 2026 – Present

- Performed a mechanistic audit of a state-of-the-art MoE-LoRA continual instruction-tuning method for large vision-language models (LLaVA-1.5-7B) across 14 multimodal tasks: showed that task identification is driven by CLIP prototype anchors rather than the learned router, and traced much of the observed forgetting on caption tasks to routing confusion between highly similar task anchors.
- Designed a **training-free, inference-only** per-sample routing purification—no retraining and negligible run-time overhead—that restores captioning from 77.6 to 134.0 CIDEr (≈ the oracle ceiling; benchmark mean +2.2) and ports inference-only to two other published methods (+74.2 / +54.3 CIDEr); validated on the **UCIT** and **CoIN** benchmarks at full official test-split scale (up to N = 214k); manuscript in preparation as a position paper for **ICML 2027**.

SLARC: Shrinkage-Controlled Alignment & Regularized Classification Jan. 2026 – Sep. 2026

- Kept cached class statistics accurate as a LoRA-adapted ViT keeps changing its features without old images: a **low-rank affine correction** in a displacement-aware subspace, scaled by a **split-fit shrinkage** rule (agreement between fits on disjoint image halves), plus a **regularized Gaussian head** that mixes class-specific and pooled covariances with count-adaptive weights; head hyperparameters are selected from scores archived when each class is introduced.
- Plug-and-play on three hosts (E^2-LoRA, SLCA, TUNA): improved E^2-LoRA final accuracy by 0.70–4.08 points on CIFAR-100, ImageNet-R, ImageNet-A, and Cars-196 (ten-task), gains persisting at 50 tasks; under review at **ICLR 2027**.

Feature-Space Alignment for Exemplar-Free Continual Learning May. 2025 – Mar. 2026

- *Geometry-anchored transport:* modeled the old-to-new feature mapping as a **closed-form linear transform** (solved in closed form, no training) plus a small learned residual for the nonlinear remainder, keeping cached old-class statistics accurate; improved final accuracy over the prior state of the art (DPCR) by 1.8–7.4 points across CIFAR-100, TinyImageNet, and ImageNet-100—first-author paper at **ECCV 2026**.
- *Bidirectional alignment with cycle consistency:* learned forward (old→new) and backward (new→old) feature maps tied by a cycle-consistency constraint, with a supporting theoretical guarantee; improved over the prior method (AdaGauss) by 1.2–3.8 points on the same benchmarks—first-author paper at **ICLR 2026**.

Large-Scale Instance-Segmentation Benchmark & Model Gallery Aug. 2022 – Jul. 2024

- Benchmarked 18 instance-segmentation methods across ∼**3M annotated instances** (18 imaging domains) from two public microscopy datasets, building the full **data-preprocessing, distributed-training, and quantitative-evaluation pipeline** on a multi-GPU cluster.
- Showed that general-purpose attention/foundation-style backbones (e.g., Swin Transformer, ResNeSt) outperform domain-specific tools, with the right input channels raising accuracy up to 312%; open-sourced **SegGal**, a gallery of 138 pretrained models; co-first author, **Briefings in Bioinformatics** (2024).

Neural-Network Security on ML Accelerators Jan. 2023 – Apr. 2023

- Studied how logic-locking mechanisms on machine-learning **hardware accelerators** can be exploited to inject a neural-Trojan attack, characterizing the attack surface and its impact on model inference under real deployment constraints; first-author paper at **GLSVLSI** (2023).

Selected Publications & Service

- **H. Xu** and B. Krawczyk, *Geometry-Anchored Transport Framework for Exemplar-Free Class-Incremental Learning.* In **ECCV**, 2026.
- **H. Xu** and B. Krawczyk, *Two-Way Is Better Than One: Bidirectional Alignment with Cycle Consistency for Exemplar-Free Class-Incremental Learning.* In **ICLR**, 2026.
- **H. Xu** and B. Krawczyk, *Revisiting Prototype Rehearsal for Exemplar-Free Continual Learning: Manifold-Aware Boundary Sampling with Adaptive Class-Balanced Loss.* In **CVPR Findings**, 2026.
- Y. Wang*, J. Zhao*, **H. Xu***, et al., *A Systematic Evaluation of Computational Methods for Cell Segmentation.* **Briefings in Bioinformatics**, 2024. [SegGal] *Equal contribution.
- A. Naghizadeh, **H. Xu**, et al., *Semantic Aware Data Augmentation for Cell Nuclei Microscopical Images with Artificial Neural Networks.* In **ICCV**, 2021.
- **H. Xu**, D. Liu, C. Merkel, and M. Zuzak, *Exploiting Logic Locking for a Neural Trojan Attack on Machine Learning Accelerators.* In **GLSVLSI**, 2023.
- **Service:** Gold Reviewer Award, **ICML 2026**; Reviewer for **ICLR 2027**, **AAAI 2027**, **NeurIPS 2026**, **AAAI 2026**, *Image and Vision Computing*, and *Frontiers in Immunology*.

Skills

- Programming Languages
  - Python
  - C++
  - SQL
  - Bash, R
- Research Areas
  - Computer Vision
  - Multimodal LLMs / VLMs
  - Continual Learning
  - Segmentation & Perception
- Models & Methods
  - Vision Foundation Models
  - VLMs (LLaVA, Qwen, BLIP, CLIP)
  - Transformers, GANs
  - PEFT (LoRA, MoE-LoRA)
- Engineering
  - PyTorch, OpenCV
  - HuggingFace Transformers
  - Linux / CUDA / Git
  - Distributed Training & Large-Scale Data Pipelines